

Hybrid Gaussians for Robust Open-Vocabulary 3D Segmentation with Multi-View Object Association and Boundary Refinement

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Abstract. Open-vocabulary 3D segmentation aims to localize objects in 3D scenes from free-form text queries, providing a flexible interface for language-guided scene understanding. Recent progress in vision-language models and neural scene representations has advanced this area by lifting 2D cues into 3D and enriching NeRF- or 3DGS-based scene representations with semantic or identity information. However, robust object-level segmentation remains challenging in real image sequences, where incomplete or noisy 2D supervision can destabilize multi-view identity assignment, while full-scene semantic learning can weaken object-level discriminability. To address these issues, we introduce Hybrid Gaussians, a unified 3D representation that jointly models object association and language-aligned semantics, along with a Multi-View Object Association mechanism that combines Observation Fusion and Semantic Contrastive Learning. We further introduce Boundary Reconstruction Optimization to explicitly improve contour quality by refining local boundary structure. Experiments on LERF and 3D-OVS demonstrate strong quantitative and qualitative performance. In particular, our method achieves 59.1% mIoU on LERF, yielding a 13.43% relative gain over the baseline. These results show that jointly optimizing object association and boundary-aware refinement substantially improves open-vocabulary 3D segmentation.

Keywords: Open-Vocabulary Segmentation · 3D Gaussian Splatting · Object-Aware Supervision · Multi-View Consistency

1 Introduction

Open-vocabulary 3D segmentation aims to localize and segment objects in 3D scenes from free-form text queries rather than a predefined label set. This capability is increasingly important for embodied perception and robotic interaction, as well as for immersive media and language-guided scene understanding [6, 13, 41], where systems are expected to interpret open-ended user queries in complex visual environments. In such settings, a practical approach should not only align 3D representations with language, but also maintain object-level consistency across views and recover accurate object reconstruction under challenging conditions.

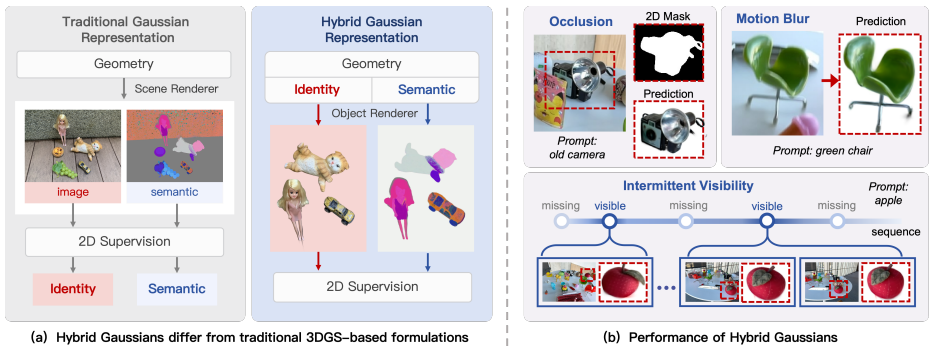


Fig. 1: (a) Comparison between traditional and Hybrid Gaussian formulations. Hybrid Gaussians couple object assignment with language-aligned semantic representation within object-conditioned primitives, rather than treating identity and semantic features as independently appended attributes. **(b) Hybrid Gaussians improve object-level open-vocabulary 3D segmentation** in challenging scenes with occlusion, motion blur, and incomplete observations, yielding more consistent object support and cleaner segmentation results.

Recent advances in vision-language models and 3D scene representations have broadened open-vocabulary 3D segmentation. Early point-based methods lifted 2D foundation-model supervision into 3D for open-vocabulary semantic reasoning [25, 33, 40]. NeRF [23] and 3D Gaussian Splatting (3DGS) [9] subsequently provided rendering-friendly scene representations for language-guided perception. Methods such as LERF, 3D-OVS, Feature-3DGS, LangSplat, and LEGaussians enrich these representations with language-aligned features for text-driven querying and segmentation [10, 19, 27, 31, 44]. Other works introduce identity-aware grouping, object-specific Gaussian sets, or object-centric anchors to improve object-level modeling [21, 42, 45]. More recent methods combine instance and semantic cues through instance-aware grouping, identity-guided feature learning, cross-modal semantic alignment, instance-to-language mapping, or multi-view semantic aggregation [2, 7, 16, 38, 46].

Despite this progress, existing open-vocabulary 3D segmentation methods still struggle to maintain consistent segmentation when lifted 2D cues are incomplete, noisy, or intermittently available across views. To address these issues, we propose Hybrid Gaussians, a unified 3D representation in which each Gaussian carries both an object identity variable and a learnable language-aligned semantic embedding. As shown in Figure 1(a), prior Gaussian-based methods typically treat identity and semantic cues as separate learned signals or auxiliary attributes for downstream segmentation. In contrast, Hybrid Gaussians integrate object identity and language-aligned semantics into a unified rendering representation, enabling object-guided supervision, as shown in Figure 1(b).

Building on this hybrid representation, we further propose a Multi-view Object Association mechanism composed of two complementary components: Ob-

064 servation Fusion and Semantic Contrastive Learning. Observation Fusion aggregates 064
 065 object-specific evidence across views to update Gaussian identities from 065
 066 fused multi-view support rather than isolated single-view observations, thereby 066
 067 providing more stable object support for 3D learning. On top of this, Semantic 067
 068 Contrastive Learning performs object-conditioned semantic rendering and 068
 069 cross-view contrastive alignment, reducing cross-object feature interference and 069
 070 producing more discriminative object expressiveness for open-vocabulary query- 070
 071 ing. Together, these two components couple stable multi-view identity support 071
 072 with object-specific semantic learning, providing a more reliable basis for robust 072
 073 open-vocabulary 3D segmentation. 073

074 To further improve spatial precision, we additionally introduce Boundary 074
 075 Reconstruction Optimization, which explicitly constrains contour reconstruction 075
 076 near object boundaries. This boundary-focused refinement aligns rendered ob- 076
 077 ject gradients with masked image observations in the contour neighborhood, 077
 078 improving local geometry and appearance consistency and leading to sharper 078
 079 segmentation results. In summary, the contributions of our work are as follows: 079

- 080 – We introduce a hybrid Gaussian formulation that supports object-conditioned 080
 081 rendering and learning by jointly maintaining identity assignment states and 081
 082 language-aligned semantics on the same primitives. 082
- 083 – We identify a key challenge of open-vocabulary 3D segmentation as the cou- 083
 084 pling between inconsistent object support across views and weak object-level 084
 085 semantic discrimination, and address it via a multi-view object association 085
 086 scheme consisting of observation fusion and object-conditioned semantic con- 086
 087 trastive learning. 087
- 088 – We further refine contour quality with a lightweight boundary reconstruction 088
 089 objective. 089
- 090 – Experiments on the LERF and 3D-OVS benchmarks demonstrate the effec- 090
 091 tiveness of the proposed method, with strong qualitative and quantitative 091
 092 performance. 092

093 2 Related Work 093

094 2.1 Segmentation on Explicit 3D Primitives 094

095 Early 3D segmentation methods mainly operated on explicit primitives such 095
 096 as point clouds and volumetric grids [26, 34]. PointGroup [8], HAIS [3], and 096
 097 SoftGroup [35] group points into object instances based on geometric consistency, 097
 098 whereas Mask3D [30] and ISBNNet [24] formulate 3D instance segmentation as 098
 099 mask prediction with object queries. 099

100 These methods usually rely on category-specific 3D annotations and are 100
 101 therefore not well suited to zero-shot open-vocabulary recognition. A common so- 101
 102 lution is to leverage 2D foundation models, including CLIP-LSeg [15], SAM [12], 102
 103 and SAM2 [29], as scalable supervision for 3D segmentation [37]. OpenScene [25] 103
 104 learns dense CLIP-aligned point features for arbitrary-category semantic seg- 104
 105 mentation, while OpenMask3D [33] combines class-agnostic 3D mask propos- 105
 106 als with aggregated multi-view CLIP features. Open-Fusion [40] further enables 106

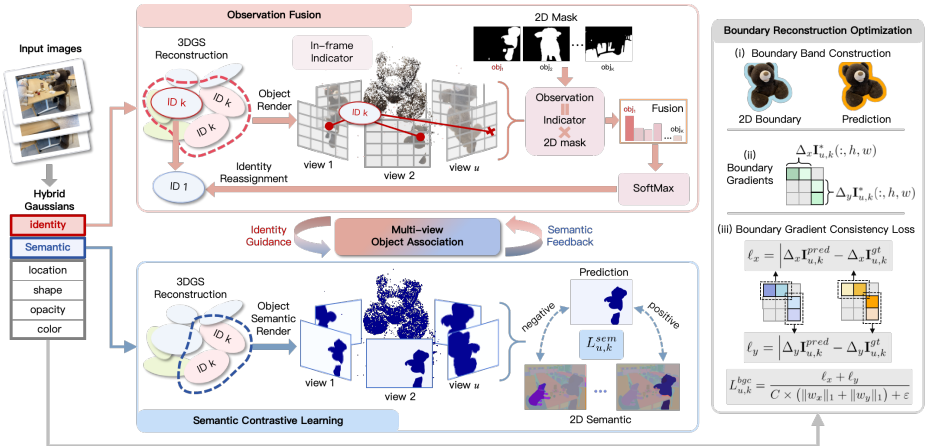


Fig. 2: Training pipeline of our method. We jointly optimize hybrid 3D Gaussians through multi-view observation fusion for identity reassignment, semantic contrastive learning for object-aware language discrimination, and boundary reconstruction optimization for sharper local details, enabling robust open-vocabulary 3D segmentation.

annotation-free 3D segmentation without additional 3D training. More broadly, many subsequent methods lift 2D predictions into 3D.

2.2 Segmentation on Neural Representations

Recent studies have also extended language-guided learning to neural scene representations, initially using NeRF [23]. LERF [10] aligns rendered features with CLIP to construct a dense language-aware radiance field; 3D-OVS [19] distills open-vocabulary semantics into NeRF using scene-level text supervision; and Semantic-NeRF [36] introduces a dedicated semantic prediction head.

Following the emergence of 3DGS [9], semantic embeddings were incorporated into Gaussians for efficient semantic-aware rendering. Feature-3DGS [44] distills dense 2D feature fields into 3DGS, while LangSplat [27] integrates CLIP-aligned features into Gaussians to form a 3D language field and later extends this formulation to dynamic scenes [17]. LEGaussians [31] instead uses compact quantized semantic codes to avoid storing high-dimensional CLIP features. Other works introduce identity embeddings: Gaussian Grouping [42] assigns each Gaussian a learnable identity embedding for view-consistent segmentation, and Segment-then-Splat [21] initializes identity-specific Gaussian sets from multi-view masks and imposes set-wise reconstruction constraints. Recent methods increasingly combine instance and language cues. OpenGaussian [38] learns 3D-consistent instance features and associates clustered instances with multi-view CLIP features. ILGS [7] jointly learns identity and language embeddings with identity-aware semantic consistency, while Seg-Splat [21] reconstructs separate object-specific Gaussian sets before assigning object-level CLIP embeddings.

ObjectGS [45] initializes object-aware anchors from multi-view voted IDs and uses discrete Gaussian semantics. COS3D [46] links instance and language fields through instance-to-language mapping, whereas LaGa [2] aggregates multi-view semantic descriptors after object decomposition.

In contrast, our method updates per-Gaussian object assignments during optimization by fusing multi-view mask evidence. These assignments define $\mathcal{G}_k$ and determine the Gaussian support for object-conditioned semantic rendering, appearance reconstruction, and boundary optimization.

3 Method

In this section, we first derive object-specific 2D supervision from propagated masks and image features. We then build Hybrid Gaussians as the unified scene representation and optimize them through multi-view object association and boundary reconstruction, as shown in Figure 2.

3.1 Preliminary: 2D Propagation

Given ground-truth images captured from a set of views $\mathcal{U}$:

$$\mathcal{I} = \{I_u^{gt}\}_{u \in \mathcal{U}}, \quad (1)$$

we consider K foreground objects and one background label, yielding $K+1$ labels. We use SAM [12] to initialize object masks and SAM2 [29] to propagate them across views. For object k in view u , the propagated binary mask is:

$$M_{u,k}^{gt} = \mathbb{I}(\zeta(I_u^{gt}, k) \geq \tau_1), \quad M_{u,k}^{gt} \in \{0, 1\}^{H \times W}, \quad (2)$$

where ζ denotes SAM2, τ_1 is a confidence threshold, and $\mathbb{I}(\cdot)$ is the indicator function. The corresponding object-specific RGB target is:

$$I_{u,k}^{gt} = I_u^{gt} \odot M_{u,k}^{gt}, \quad I_{u,k}^{gt} \in \mathbb{R}^{C \times H \times W}. \quad (3)$$

Beyond object-conditioned photometric supervision, we also introduce a semantic feature learning process as an additional supervisory signal. We extract a semantic feature map using a frozen 2D CLIP image encoder $\Phi_i(\cdot)$ [28]:

$$S_{u,k}^{gt} = \Phi_i(I_u^{gt}) \odot M_{u,k}^{gt}, \quad S_{u,k}^{gt} \in \mathbb{R}^{D \times H \times W}, \quad (4)$$

The object-cropped ground-truth $M_{u,k}^{gt}$, $I_{u,k}^{gt}$, and $S_{u,k}^{gt}$ serve as supervision signals for the subsequent object-specific learning.

3.2 Hybrid Gaussians

Following basic 3DGS [9], we represent a scene by a set of N Gaussian primitives reconstructed from the image set $\mathcal{I}$ in Equation (1):

$$\mathcal{G} = \{g_n = (l_n, \Sigma_n, \delta_n, c_n)\}_{n=1}^N, \quad (5)$$

where each Gaussian is parameterized by its center $l_n \in \mathbb{R}^3$, covariance $\Sigma_n \in \mathbb{R}^{3 \times 3}$, opacity $\delta_n \in \mathbb{R}$, and color $c_n \in \mathbb{R}^3$.

To enable open-vocabulary 3D segmentation, we extend this basic representation into a hybrid one by equipping each Gaussian with both an identity variable and a semantic feature:

$$\mathcal{G} = \{g_n = (l_n, \Sigma_n, \delta_n, c_n, z_n, f_n)\}_{n=1}^N. \quad (6)$$

where $z_n \in \{0, 1, \dots, K\}$ denotes the object-level segmentation identity and $f_n \in \mathbb{R}^D$ is a learnable language-aligned semantic embedding. This design yields Hybrid Gaussians: by attaching semantic and identity attributes directly to the same primitives, the representation establishes a unified 3D carrier for segmentation and reconstruction.

Unlike prior Gaussian-based representations that mainly attach identity or semantic features as auxiliary attributes [7, 16, 38], our hybrid formulation explicitly models object association in each Gaussian through jointly optimized identity and semantic representations, enabling object-conditioned rendering and learning. Building on this foundation, multi-view object association aggregates 2D cues across views to progressively stabilize identity assignments and improve semantic representation learning under ambiguous or intermittent supervision. As a result, Hybrid Gaussians provide a unified representation that supports robust segmentation while still allowing geometry attributes in Equation (5) to be further refined by reconstruction and boundary-aware optimization.

3.3 Multi-view Object Association

Observation Fusion Prior 3DGS methods associate Gaussians with object identities through learned encodings, voted IDs, or object-specific sets [21, 42, 45]. Such associations can be unreliable under incomplete or ambiguous views. We therefore maintain a discrete assignment z_n and update it during training by fusing multi-view mask support. Unlike ObjectGS [45], which transfers voted IDs to initialize object-aware points and anchors, our updated assignments define the support for object-conditioned rendering and semantic supervision.

To extract object-specific observations for identity reassignment, we first render each object separately according to the current identity assignments. Formally, we define the object-specific subset

$$\mathcal{G}_k \triangleq \{g_n \in \mathcal{G} \mid z_n = k\}. \quad (7)$$

Thus, $\mathcal{G}_k$ is not merely a post-hoc grouping of Gaussians: it is the conditioning variable that determines the support of all object-specific rendering and supervision operators in our framework. We render $\mathcal{G}_k$ with an object renderer $\mathcal{R}_{\text{obj}}$ as in [21]:

$$I_{u,k}^{\text{pred}} = \mathcal{R}_{\text{obj}}(\mathcal{G}_k, u), \quad I_{u,k}^{\text{pred}} \in \mathbb{R}^{C \times H \times W}. \quad (8)$$

To determine whether a Gaussian contributes a valid projected observation to the object-specific prediction $I_{u,k}^{\text{pred}}$ under view u , we follow the visibility-aware

rendering and frustum-culling practice in 3DGS-based renderers [9, 14]. Given the camera intrinsics $K_u \in \mathbb{R}^{3 \times 3}$ and extrinsics (R_u, t_u) , we first transform the Gaussian center l_n into the camera coordinate system:

$$P_{n,u} = R_u l_n + t_u = (x_{n,u}, y_{n,u}, d_{n,u})^\top. \quad (9)$$

The corresponding pixel coordinate $(x_{n,u}^p, y_{n,u}^p)$ is obtained by projection:

$$(\tilde{x}_{n,u}, \tilde{y}_{n,u}, \tilde{d}_{n,u})^\top = K_u P_{n,u}, \quad x_{n,u}^p = \frac{\tilde{x}_{n,u}}{\tilde{d}_{n,u}}, \quad y_{n,u}^p = \frac{\tilde{y}_{n,u}}{\tilde{d}_{n,u}}. \quad (10)$$

We then define an in-frame indicator $V_{n,u} \in \{0, 1\}$ to determine whether Gaussian g_n yields a valid projected observation:

$$V_{n,u} = \mathbb{I}(d_{n,u} > 0) \cdot \mathbb{I}(0 \leq x_{n,u}^p < W) \cdot \mathbb{I}(0 \leq y_{n,u}^p < H). \quad (11)$$

Here, $V_{n,u} = 1$ only if the Gaussian lies in front of the camera ($d_{n,u} > 0$) and its projected center falls inside the image domain. This indicator determines whether the mask cue from view u is eligible to contribute to the identity update of Gaussian g_n .

Based on this validity criterion, we define the observation of Gaussian g_n from view u as the object-specific mask support at its projected 2D location. Concretely, for each candidate label $k \in \{0, 1, \dots, K\}$, we query the 2D object mask $M_{u,k}^{gt}$ at the projected location $(x_{n,u}^p, y_{n,u}^p)$ and define

$$o_{n,u,k} = V_{n,u} M_{u,k}^{gt}(x_{n,u}^p, y_{n,u}^p). \quad (12)$$

The scalar $o_{n,u,k}$ indicates whether Gaussian g_n is both validly projected in view u and supported by the mask of object k . Therefore, $\{o_{n,u,k}\}_{k=0}^K$ constitutes the view-wise object observation of Gaussian g_n under view u .

Since each $o_{n,u,k}$ is derived from a single view and thus provides only partial evidence, we perform observation fusion by aggregating these object-specific observations across views. For each Gaussian g_n and candidate label k , the fused observation is computed as

$$O_{n,k} = \sum_{u \in \mathcal{U}} w_u o_{n,u,k}, \quad (13)$$

where $w_u > 0$ denotes a learnable weight. Collecting the fused observations over all candidate labels yields the latent logit vector

$$\ell_n = [O_{n,0}, O_{n,1}, \dots, O_{n,K}] \in \mathbb{R}^{K+1}. \quad (14)$$

The updated identity assignment is then obtained by normalizing the fused logits and selecting the most supported label:

$$q_n = \text{softmax}(\ell_n), \quad z_n = \arg \max_{k \in \{0, \dots, K\}} q_{n,k}. \quad (15)$$

In this way, Multi-view Observation Fusion consolidates view-dependent mask observations into reliable identity evidence for each Gaussian, thereby enabling more stable identity reassignment under ambiguous or intermittent supervision.

Semantic Contrastive Learning While Multi-view Observation Fusion stabilizes identity assignments, it does not resolve the issue of weak semantic expressiveness. If semantic supervision is applied at the scene level, features from different objects can still be corrupted by 2D-to-3D lifting noise [38]. To address this limitation, we propose an object-specific semantic rendering strategy, which conditions semantic rendering on the subset $\mathcal{G}_k \subset \mathcal{G}$. This design couples semantic learning with the object support established by identity reassignment, thereby reducing cross-object feature mixing and providing a cleaner supervisory signal for semantic representation learning. Formally, given the semantic embeddings $\{f_n\}_{g_n \in \mathcal{G}_k}$ of the selected Gaussians, we render an object-specific semantic feature map in view u through the semantic renderer $\mathcal{R}_{\text{sem}}$:

$$S_{u,k}^{\text{pred}} = \mathcal{R}_{\text{sem}}(\mathcal{G}_k, \{f_n\}_{g_n \in \mathcal{G}_k}, u), \quad S_{u,k}^{\text{pred}} \in \mathbb{R}^{D \times H \times W}. \quad (16)$$

This object-conditioned rendering restricts the semantic prediction to the Gaussian subset associated with object k , thereby reducing interference from irrelevant scene regions.

We then restrict both the predicted and reference semantic maps to the reference object region defined by Equation (2), so that semantic alignment is performed only on the valid object support:

$$F_{u,k}^* = S_{u,k}^* \odot M_{u,k}^{\text{gt}}, \quad F_{u,k}^* \in \mathbb{R}^{D \times H \times W}, \quad * \in \{\text{pred}, \text{gt}\}. \quad (17)$$

For each anchor feature map $F_{u,k}^{\text{pred}}$, we construct multi-view contrastive pairs according to object identity. The feature map of the same object k from another view $v \neq u$ is treated as a positive sample, denoted by $F_{v,k}^{\text{gt}}$, while feature maps of different objects in the batch are treated as negative samples, i.e., $\{F_{v,k'}^{\text{gt}} \mid k' \neq k\}$. The resulting semantic InfoNCE loss [11] is defined as:

$$L_{u,k}^{\text{sem}} = -\log \frac{\exp\left(\text{sim}\left(\hat{F}_{u,k}^{\text{pred}}, \hat{F}_{v,k}^{\text{gt}}\right) / \tau_2\right)}{\sum_{k'=0}^K \exp\left(\text{sim}\left(\hat{F}_{u,k}^{\text{pred}}, \hat{F}_{v,k'}^{\text{gt}}\right) / \tau_2\right)}, \quad (18)$$

where $\hat{F}$ denotes the ℓ_2 -normalized flattened feature map, $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and τ_2 is a temperature hyper-parameter. Minimizing Equation (18) encourages the learnable semantic features $\{f_n\}$ to produce object representations that are consistent for the same object across views and discriminative among different objects, thereby improving semantic separability and reducing context leakage from irrelevant regions.

Together, Observation Fusion and Semantic Contrastive Learning form a complementary multi-view object association mechanism: Observation Fusion first stabilizes object identity assignment, thereby providing reliable object support for the subsequent semantic supervision; Semantic Contrastive Learning then improves the discriminability of the associated object, which in turn strengthens the robustness of object-level representation across views. As a result, the Hybrid Gaussian representation supports both consistent identity estimation and semantically coherent object embeddings.

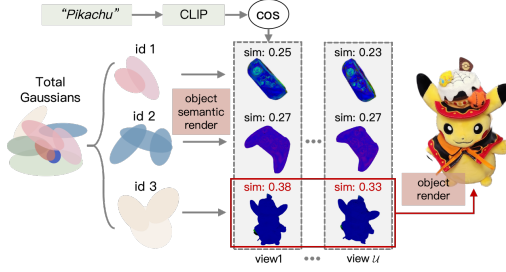


Fig. 3: Open-vocabulary segmentation. For a given text query, we select the object with the highest cosine similarity for final segmentation.

3.4 Boundary Reconstruction Optimization

With reliable object association, we refine Gaussian geometry through object-level appearance reconstruction. Specifically, we define an object-level appearance reconstruction loss between the prediction in Equation (8) and the masked ground-truth in Equation (3), where $\|\cdot\|_1$ denotes L_1 loss:

$$L_{u,k}^{appear} = (1 - \lambda) \left\| I_{u,k}^{pred} - I_{u,k}^{gt} \right\|_1 + \lambda \left(1 - \text{SSIM} \left(I_{u,k}^{pred}, I_{u,k}^{gt} \right) \right). \quad (19)$$

This loss provides object-level photometric supervision, but it does not explicitly emphasize contour quality. In practice, boundary regions remain particularly vulnerable to semantic ambiguity [43] and geometric misalignment [4], which often leads to blurred or inaccurate object outlines. To address this issue, we introduce Boundary Reconstruction Optimization, which explicitly constrains boundary reconstruction near object contours.

We first construct a predicted boundary band $B_{u,k}^{pred} \in \{0, 1\}^{H \times W}$ from the rendered object image using a dilation-erosion contrast [32] with channel-wise max pooling $\mathcal{P}_{max}$:

$$\text{dil}(I_{u,k}^{pred}) = \mathcal{P}_{max}(I_{u,k}^{pred}), \quad \text{er}(I_{u,k}^{pred}) = 1 - \mathcal{P}_{max}(1 - I_{u,k}^{pred}), \quad (20)$$

$$B_{u,k}^{pred}(h, w) = \begin{cases} 1, & \text{if } \text{dil}(I_{u,k}^{pred})(h, w) - \text{er}(I_{u,k}^{pred})(h, w) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

This band identifies the contour neighborhood of the current prediction and restricts the boundary supervision to regions most relevant to shape refinement.

We then compute forward finite differences for both the rendered prediction and the masked ground-truth image, where $*$ denotes either *pred* or *gt*:

$$\Delta_x I_{u,k}^*(:, h, w) = I_{u,k}^*(:, h, w + 1) - I_{u,k}^*(:, h, w), \quad w = 1, \dots, W - 1, \quad (22)$$

$$\Delta_y I_{u,k}^*(:, h, w) = I_{u,k}^*(:, h + 1, w) - I_{u,k}^*(:, h, w), \quad h = 1, \dots, H - 1. \quad (23)$$

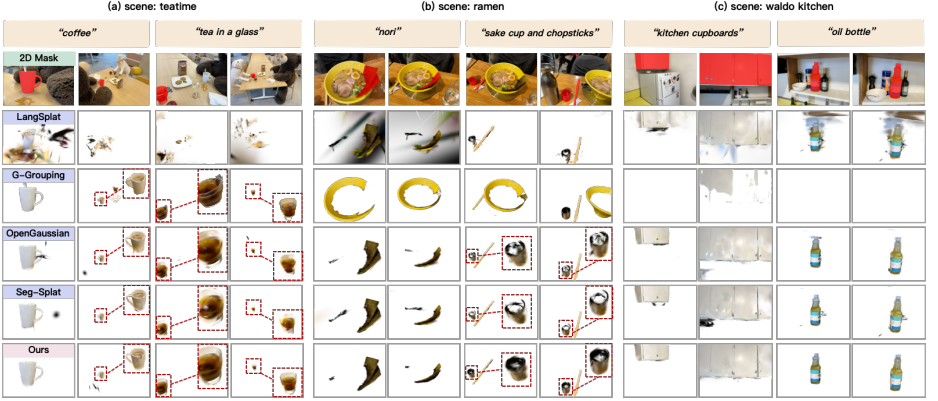


Fig. 4: Qualitative comparison. For each text query, we visualize the corresponding object segmentation results from different views. Compared with prior methods, our method achieves more accurate object localization, cleaner segmentation boundaries, and stronger cross-view consistency. Gaussian Grouping and Segment-then-Splat are abbreviated as G-Grouping and Seg-Splat, respectively.

To align the boundary weights with the discrete gradient tensors, we slice the predicted band to obtain

$$w_x = B_{u,k}^{pred}(:, 2:W), \quad w_y = B_{u,k}^{pred}(2:H, :), \quad (24)$$

where $w_x \in \{0, 1\}^{H \times (W-1)}$ and $w_y \in \{0, 1\}^{(H-1) \times W}$ weight horizontal and vertical gradients, respectively.

Based on these weights, we define a boundary gradient consistency loss that penalizes gradient discrepancies only within the predicted contour neighborhood:

$$\ell_x = \sum_{c=1}^C \sum_{h=1}^H \sum_{w=1}^{W-1} w_x(h, w) \left| \Delta_x I_{u,k}^{pred}(c, h, w) - \Delta_x I_{u,k}^{gt}(c, h, w) \right|, \quad (25)$$

$$\ell_y = \sum_{c=1}^C \sum_{h=1}^{H-1} \sum_{w=1}^W w_y(h, w) \left| \Delta_y I_{u,k}^{pred}(c, h, w) - \Delta_y I_{u,k}^{gt}(c, h, w) \right|, \quad (26)$$

$$L_{u,k}^{bgc} = \frac{\ell_x + \ell_y}{C \times (\|w_x\|_1 + \|w_y\|_1) + \varepsilon}, \quad (27)$$

where ε is a small constant for numerical stability. By enforcing consistency between predicted and ground-truth image gradients along the contour neighborhood, this loss encourages more spatially accurate object boundaries.

Finally, the overall training objective combines semantic contrastive, boundary refinement, and appearance losses:

$$L = \sum_{u \in \mathcal{U}} \sum_{k=0}^K \left(\lambda_{sem} L_{u,k}^{sem} + \lambda_{bgc} L_{u,k}^{bgc} + \lambda_{appear} L_{u,k}^{appear} \right). \quad (28)$$

Table 1: Quantitative comparisons of 3D segmentation on the LERF and 3D-OVS datasets with mIoU scores (%). In all tables, the best results are shown in bold and the second-best results are underlined.

Methods	LERF					3D-OVS					
	<i>figurines</i>	<i>teatime</i>	<i>ramen</i>	<i>kitchen</i>	Avg.	<i>bed</i>	<i>bench</i>	<i>room</i>	<i>sofa</i>	<i>lawn</i>	Avg.
LERF [10]	38.6	45.0	28.2	37.9	37.4	73.5	53.2	46.6	27.0	73.7	54.8
ODISE [39]	-	-	-	-	-	52.6	24.1	52.5	48.3	39.8	43.5
OV-Seg [18]	-	-	-	-	-	79.8	88.9	71.4	66.1	81.2	77.5
LEGaussians [31]	40.8	60.3	46.0	39.4	46.6	52.8	61.3	51.8	42.0	59.6	53.5
G-Grouping [42]	28.8	46.8	31.1	11.6	29.6	64.5	95.6	96.4	91.3	97.0	89.0
3D-OVS [19]	44.8	56.1	28.7	39.3	42.2	89.5	89.3	92.8	74.0	88.2	86.8
OpenGaussian [38]	39.3	60.4	31.1	22.7	38.4	-	-	-	-	-	-
LangSplat [27]	44.7	<u>65.1</u>	51.2	<u>44.5</u>	51.4	92.5	94.2	94.1	90.0	96.1	93.4
Feature-3DGS [44]	58.8	40.5	43.7	-	-	83.5	90.7	84.7	86.9	93.4	87.8
Laser [22]	-	-	-	-	-	91.4	88.3	92.8	86.0	88.2	89.3
SAGA [1]	-	-	-	-	-	97.4	<u>95.4</u>	<u>96.8</u>	93.5	96.6	<u>96.0</u>
Seg-Splat [21]	49.8	63.5	<u>54.4</u>	40.7	<u>52.1</u>	95.8	95.1	78.4	<u>94.1</u>	93.0	91.3
COS3D [46]	60.0	65.1	35.9	42.1	50.8	-	-	-	-	-	-
Ours	<u>58.6</u>	66.5	57.6	53.7	59.1	<u>96.4</u>	94.9	98.0	94.5	<u>96.8</u>	96.1

3.5 Open-Vocabulary Segmentation

As shown in Figure 3, after training, each object is represented by a semantic representation derived from the associated Gaussians. Given an input text query q , we first encode it into a language feature using the CLIP text encoder $\Phi_t(\cdot)$:

$$f_q = \Phi_t(q). \quad (29)$$

For each object k , we encode its rendered semantic maps $F_{u,k}^{pred}$ over valid views $\mathcal{U}_k$ using the CLIP image encoder $\Phi_i(\cdot)$ and select the object with the highest average similarity:

$$p = \arg \max_k \frac{1}{|\mathcal{U}_k|} \sum_{u \in \mathcal{U}_k} \cos(f_q, \Phi_i(F_{u,k}^{pred})), \quad k = 0, 1, \dots, K, \quad (30)$$

where $\mathcal{U}_k$ denotes the set of valid views in which object k is observed. The object p is then segmented in the 3D reconstruction as the open-vocabulary result.

4 Experiments

4.1 Implementation Details

We evaluate our method on two 3D real-scene datasets, namely LERF [10] and 3D-OVS [19]. LERF contains complex indoor scenes and emphasizes fine-grained, text-aligned 3D scene understanding with multi-view language supervision. In contrast, 3D-OVS is a static 3D open-vocabulary segmentation dataset with

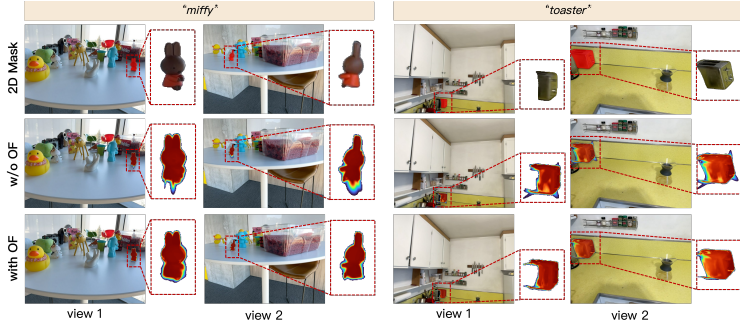


Fig. 5: Ablation study of Observation Fusion (OF). We visualize the depth maps of the queried objects across multiple views, with and without observation fusion.

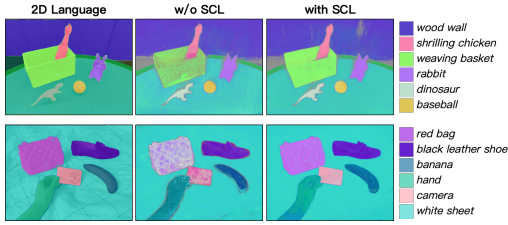


Fig. 6: Ablation study of Semantic Contrastive Learning (SCL). We compare the rendered 2D language maps without and with SCL against the 2D language supervision.

language-driven annotations for simple indoor scenes. For quantitative comparison, we report mean Intersection over Union (mIoU) [20]. For ablation studies, we further report mean pixel accuracy (mAcc) and mean boundary IoU (mBIoU).

2D object masks are generated by SAM2 with a ViT-H backbone [29]. Image and text semantic features are extracted using OpenCLIP ViT-B/16 pretrained on LAION-2B [5]. For the overall loss in Equation (28), the weights are set to $\lambda_{sem} = 0.1$, $\lambda_{bgc} = 0.1$, and $\lambda_{appear} = 1$. All experiments are conducted on an NVIDIA RTX 4090 GPU.

Additional experimental details, ablation studies, runtime analysis, limitations, and segmentation videos are provided in the supplementary material.

4.2 Qualitative Results

Figure 4 shows qualitative comparisons on representative scenes. In the “teatime” scene, all methods can roughly localize the query “coffee”, but our method produces cleaner object boundaries and less background leakage than OpenGaussian [38] and Segment-then-Splat [21]. This behavior reflects our more reliable object-level association, which better constrains the semantic response to the actual object support instead of allowing it to diffuse into surrounding regions.

The “ramen” scene further demonstrates the advantage of our method for fine-grained open-vocabulary queries. Compared with Gaussian Grouping [42], which

Table 2: Ablation study of Observation Fusion (OF) and Semantic Contrastive Learning (SCL) on the LERF in terms of mAcc (%).

OF SCL		<i>figurines</i>	<i>teatime</i>	<i>ramen</i>	<i>kitchen</i>	Avg.
-	-	95.87	96.01	95.56	91.80	94.81
✓	-	96.11	96.14	<u>96.17</u>	<u>93.00</u>	95.36
-	✓	97.26	96.67	96.11	92.02	<u>95.52</u>
✓	✓	98.09	97.77	96.23	93.34	96.36

relies on rendered 2D representations, our method better captures language-level distinctions in 3D, as illustrated by the query “sake cup and chopsticks”.

Finally, the “waldo kitchen” scene tests large-region semantic localization under weak texture cues through the query “kitchen cupboards”. Here, our method recovers more coherent cupboard regions, while previous methods tend to produce incomplete responses. LangSplat [27] mainly learns language-aligned features at the Gaussian level, but lacks sufficiently strong object-level association, making its object separation less reliable in cluttered scenes.

These qualitative results demonstrate that our method improves both object-level localization and semantic discrimination, leading to more reliable open-vocabulary segmentation across a wide range of scenes and query types.

4.3 Quantitative Results

Table 1 summarizes the quantitative comparisons on LERF [10] and 3D-OVS [19] with mIoU (%) scores. On LERF, our method achieves the best average mIoU of 59.1%, surpassing Segment-then-Splat [21] by 7.0% and LangSplat [27] by 7.7%. The gain on the “kitchen” scene is especially notable, improving over Segment-then-Splat [21] by 13.0%. These gains reflect more stable object association and sharper boundary refinement. On 3D-OVS, our method achieves the best average mIoU of 96.1%, slightly surpassing SAGA [1] at 96.0%. It obtains the best results on the “room” scene with 98.0% and the “sofa” scene with 94.5%. Compared with LangSplat [27], our method improves the average score by 2.7%. The gain on this dataset is smaller than that on LERF, because 3D-OVS usually contains fewer objects per scene and more clearly separated object regions, making the segmentation targets easier to distinguish and leaving less room for improvement from stronger object association.

Taken together, these results indicate that our method is especially effective in scenarios in which robust object association and fine-grained semantic discrimination matter most, while still maintaining strong performance on simpler benchmarks.

4.4 Ablation Studies

We first evaluate Observation Fusion (OF). As shown in Figure 5, OF yields cleaner and more coherent object structures across views. The corresponding

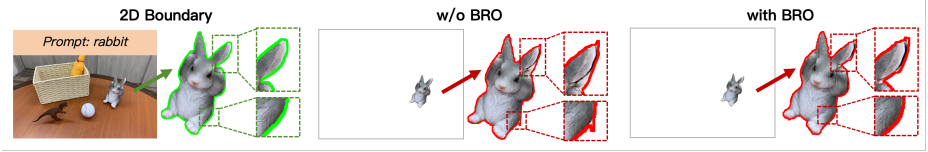


Fig. 7: Ablation study of Boundary Reconstruction Optimization (BRO): we visualize the object boundary for the queried object in variants without BRO and with BRO.

Table 3: Ablation study of BRO on the 3D-OVS in terms of mBIoU (%).

BRO	<i>bed</i>	<i>bench</i>	<i>room</i>	<i>sofa</i>	<i>lawn</i>	Avg.
-	67.18	81.26	84.22	80.74	87.48	80.18
✓	70.25	81.36	86.07	80.75	87.81	81.25

depth maps, rendered from the selected Gaussian subsets, further indicate improved spatial completeness and multi-view consistency. Quantitatively, OF increases the average mAcc from 94.81% to 95.36% in Table 2. We then evaluate Semantic Contrastive Learning (SCL). Without SCL, the rendered language maps show weaker intra-object compactness and poorer inter-object discrimination; SCL produces features that better align with the 2D language supervision, facilitating open-vocabulary segmentation in Figure 6. As reported in Table 2, SCL alone improves the average mAcc from 94.81% to 95.52%, with the largest gain of 1.39 on “figurines”. Combining OF and SCL yields the best performance, achieving the highest mAcc on all scenes and improving the average score from 94.81% to 96.36%. This outcome is expected because the two modules are complementary, their combination therefore provides both more reliable object association and separable object semantics.

We finally evaluate the proposed boundary reconstruction optimization (BRO). As shown in Figure 7, without BRO, the reconstructed contour exhibits visible local misalignment, especially in thin and high-curvature regions such as the ear and lower body of the rabbit. With BRO, the predicted boundary becomes sharper and better aligned with the 2D target, indicating improved contour fidelity. This improvement is matched quantitatively in Table 3, where BRO improves the average mBIoU (%) from 80.18% to 81.25%, with the largest gains of 3.07% on the “bed” scene and 1.85% on the “room” scene.

5 Conclusion

In this paper, we presented a robust open-vocabulary 3D segmentation framework based on Hybrid Gaussians with Multi-View Object Association. Hybrid Gaussians provide a unified representation for object-conditioned rendering and learning, Observation Fusion and Semantic Contrastive Learning improve identity stability and semantic discriminability, and Boundary Reconstruction Optimization further refines contour quality and local spatial precision.

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